DATA SCIENCE • ECONOMIC INSIGHTS

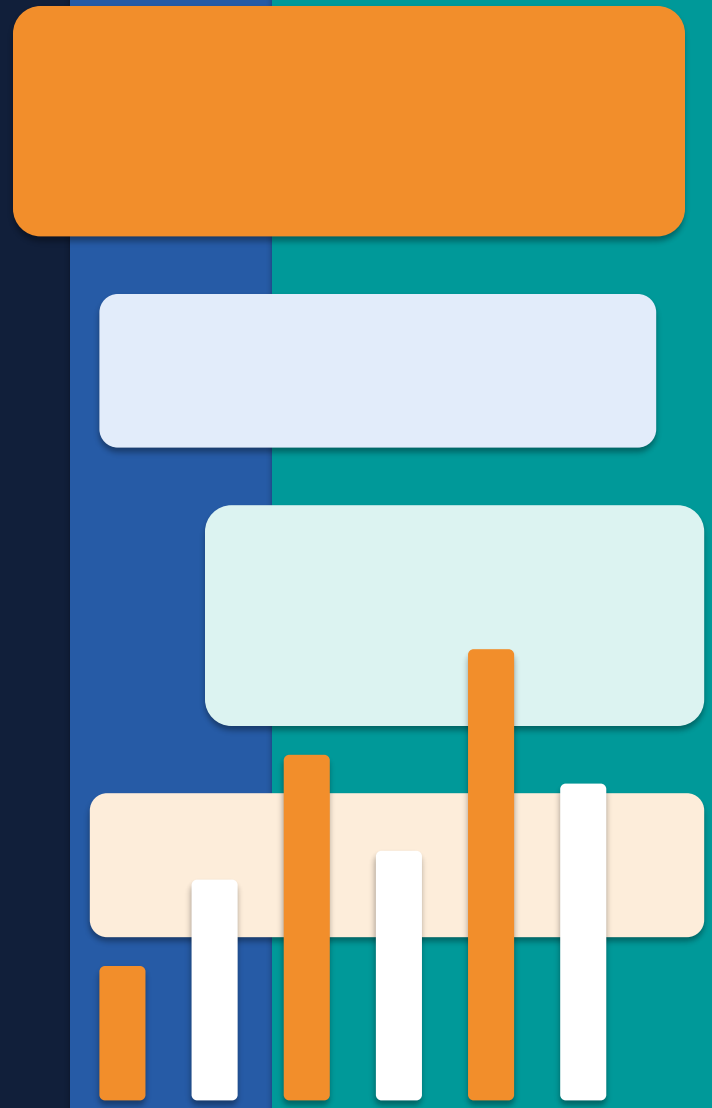
UNEMPLOYMENT ANALYSIS

Exploratory Data Analysis & Time-Series Forecasting

Prepared by John Kapungu

Python • Pandas • Scikit-learn • Statsmodels

OCTOBER 2026



Executive Summary

This project explores historical unemployment patterns and evaluates forecasting methods to support evidence-informed labor-market analysis.

01

Analyze

Clean data and explore trends, seasonality, unusual observations, and the COVID-19 period.

02

Forecast

Compare a mean baseline, persistence baseline, Random Forest, and ARIMA.

03

Inform

Translate verified patterns into practical questions for employment planning and policy.

Headline result: the persistence baseline had the lowest reported MAE and RMSE.

Project Objectives

- Prepare a reliable dataset through cleaning and preprocessing.
- Explore unemployment trends over time and investigate the COVID-19 period.
- Examine monthly or quarterly patterns and flag statistical outliers.
- Engineer historical features and evaluate forecasting models.
- Develop evidence-informed policy implications while recognizing analytical limits.

Guiding principle: conclusions must be supported by observed data.

Analytical Workflow

1 Inspect

Structure, types, missingness

2 Clean

Dates, duplicates, invalid values

3 Explore

Distributions and time trends

4 Investigate

COVID-19, seasonality, outliers

5 Engineer

Lags, rolling means, calendar features

6 Evaluate

Chronological split and error metrics

Chronological evaluation helps avoid using future observations to predict the past.

Data & Preparation

Core analytical fields

date

Timestamp used to order observations.

unemployment_rate

Target variable for trend analysis and forecasting.

Preparation checklist

- Convert dates and sort chronologically
- Check missing values and duplicates
- Validate the target variable
- Inspect observation frequency and gaps
- Save a cleaned dataset for reproducibility

Final report should specify the actual source, date range, row count, and frequency.

Exploratory Analysis

Trend analysis

Track rises, falls, and longer-term changes in unemployment.

COVID-19 period

Compare pre-pandemic, pandemic, and recovery periods where coverage allows.

Seasonality

Use monthly/quarterly summaries and heatmaps to investigate recurring patterns.

Outliers

Use IQR as a screening tool; investigate rather than automatically remove extremes.

Important: these are analytical questions; report specific patterns only after verifying the saved plots and statistics.

Forecasting Models

Mean baseline

Predicts the training-period average.

Persistence

Uses the previous observation as the next forecast.

Random Forest

Learns nonlinear relationships from lag and rolling features.

ARIMA(1,1,1)

Statistical time-series model with differencing.

Evaluation metrics: MAE and RMSE — lower values indicate smaller forecast errors.

Reported Model Performance

Lower is better for both metrics

Model	MAE ↓	RMSE ↓
Persistence Baseline	**9.049**	**10.603**
ARIMA(1,1,1)	9.738	11.342
Tuned Random Forest	9.895	11.474
Mean Baseline	10.319	11.860

Preliminary ranking: persistence baseline leads on both reported metrics.

Confirm all models use the same test dates before treating this ranking as final.

What the Results Suggest

- A simple persistence forecast was the strongest reported benchmark.
- ARIMA ranked second, while tuned Random Forest did not improve on persistence.
- Complexity alone does not guarantee stronger forecasting accuracy.
- Performance may vary across different time periods and economic conditions.

Interpretation caution: if rates are stored as percentages, MAE/RMSE are in percentage points.

Potential Policy Implications

Monitor — Use regularly updated indicators to identify rising unemployment.

Target — Use demographic and regional evidence to focus training and job-placement support.

Prepare — Consider contingency measures for future economic shocks if the analysis supports this.

Plan — Use forecasts as planning signals, not as certain predictions.

Evaluate — Measure interventions over time and use credible evaluation methods.

Recommendations should be tied to observed dataset patterns; association does not prove causation.

Limitations & Validation

- Dataset source, date range, frequency, and sample size must be documented.
- All models must be evaluated on identical test observations.
- Historical unemployment alone may omit important economic drivers.
- A single holdout period may not represent future labor-market conditions.
- COVID-19 and seasonal associations do not independently establish causation.

Next validation step: common holdout evaluation and rolling-origin backtesting.

Next Steps

- Re-evaluate all models on a shared chronological test set.
- Test exponential smoothing and alternative ARIMA configurations.
- Add relevant economic indicators where reliable data is available.
- Expand analysis by region, age, gender, or sector if supported by the data.
- Build a dashboard and publish a reproducible GitHub repository.

Deliverables: cleaned data • notebook • saved figures • README • requirements.txt • .gitignore

CONCLUSION

Start simple.
Validate rigorously.
Use data responsibly.

The persistence baseline had the lowest reported forecast errors, highlighting the value of strong benchmarks.

Thank you